

BMS_SLAVE

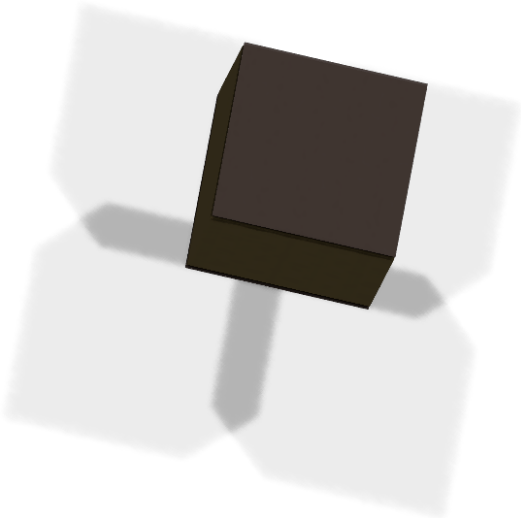
Variant: DRAFT

2026-09-25

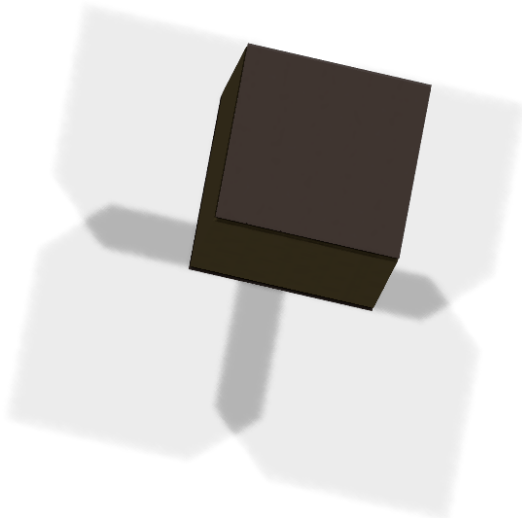
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 25-Sep-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

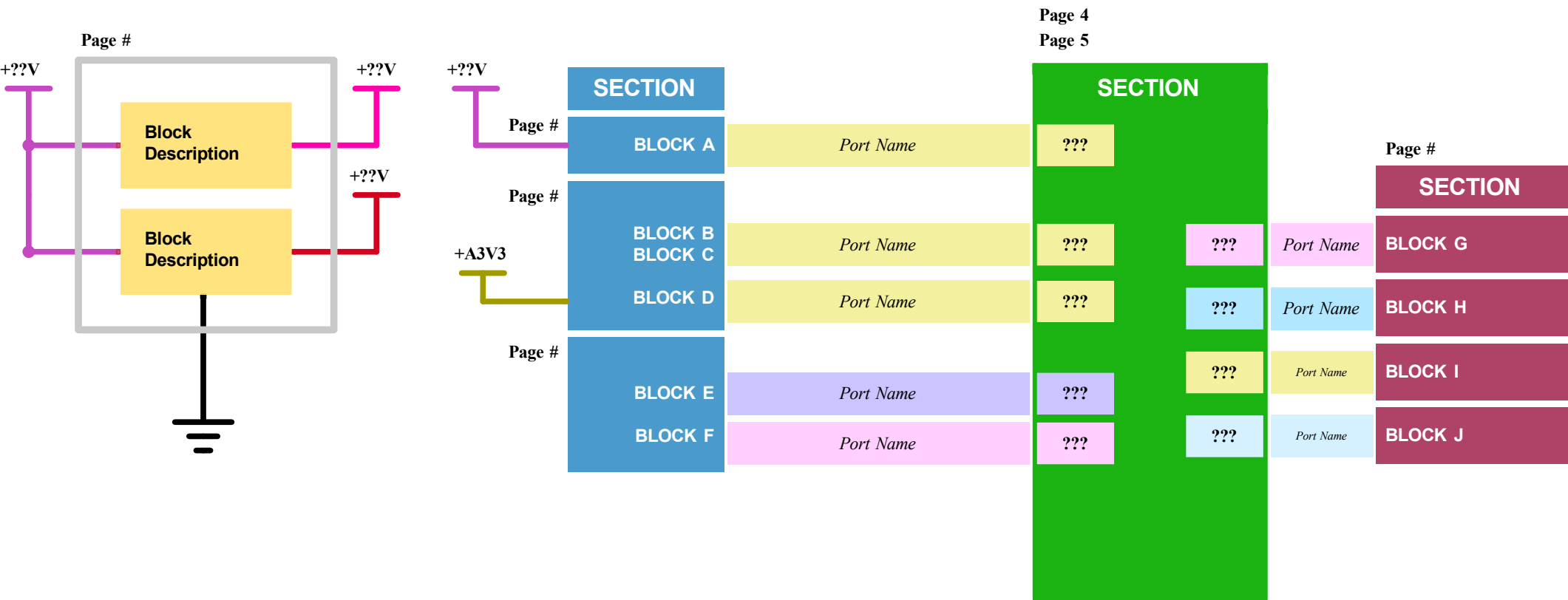
DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

	Comments:		Company: Blue Devil Motor Sports		Variant: DRAFT	Git Hash: 87d29ad
	Sheet Title:		Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Path: /		File Name: bms_slave.kicad_sch		Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)
		Reviewer:		Size: A3	Sheet: 1 of 10	

[2] Block Diagram



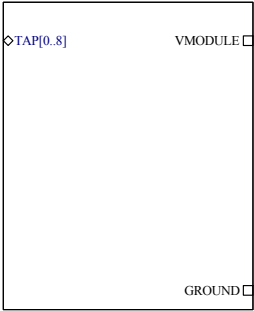
Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

	Comments:		Company:		Variant:	Git Hash:
			Blue Devil Motor Sports		DRAFT	87d29ad
			Board Name:		Project Name:	
			BMS_SLAVE		BMS Slave	
Sheet Title:		File Name:	Designer:	Date:	Revision:	
Block Diagram		Block Diagram.kicad_sch	Aidan Brzezinski	2025-01-12	+ (Unreleased)	
Sheet Path:		Reviewer:		Size:	Sheet:	
/Block Diagram/				A3	2 of 10	

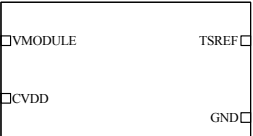
[3] Project Architecture

Page 4

Cell Sense & Balance



Power & Reference



Temperature Sense



Connectors



Daisy Chain

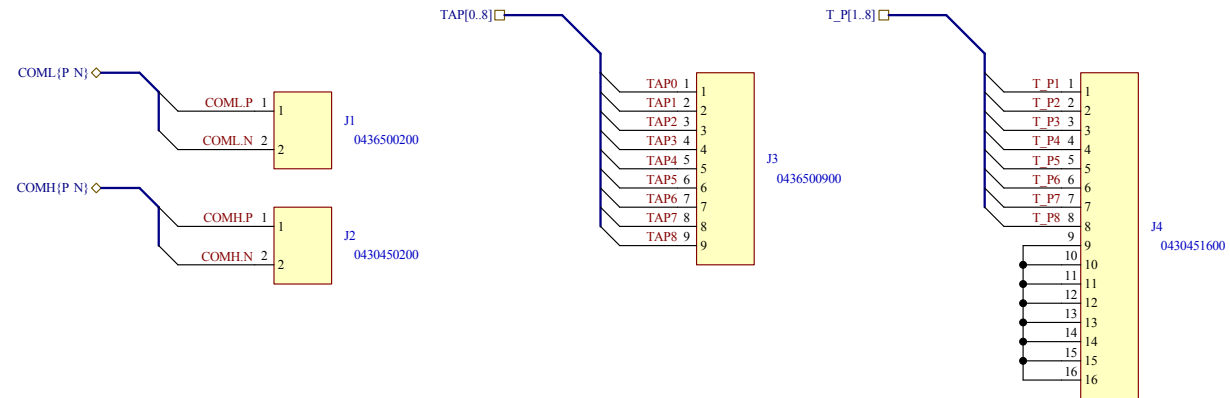


Mechanical



	Comments:		Company: Blue Devil Motor Sports		Variant: DRAFT	Git Hash: 87d29ad
			Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/			Reviewer:	Size: A3	Sheet: 3 of 10	

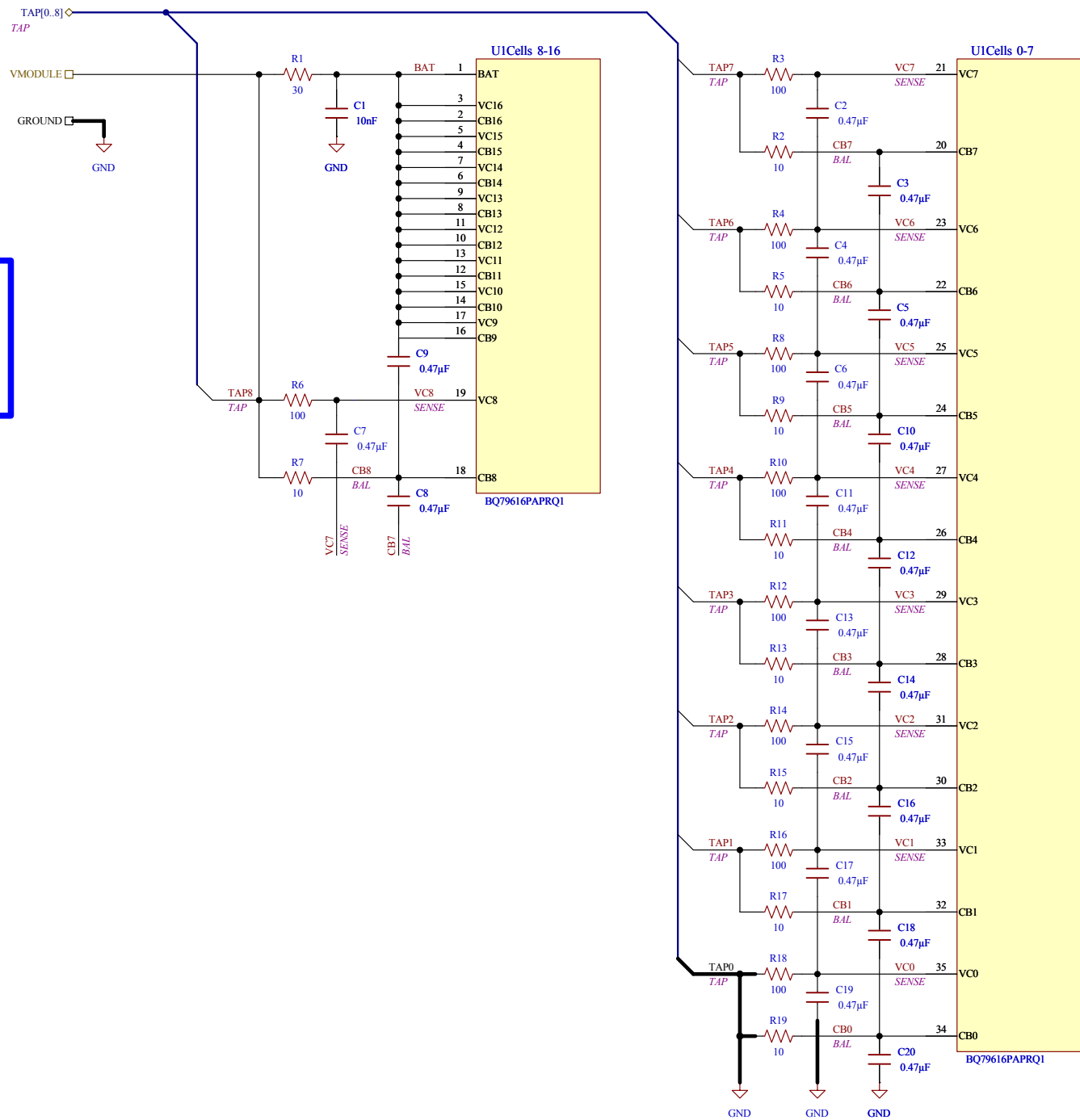
[4] Connectors



Comments:	Company:		Variant:	Git Hash:
	Board Name: BMS_SLAVE		DRAFT	87d29ad
Sheet Title: Connectors	File Name: connectors.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: A4	Sheet: 4 of 10

[5] Cell Sense & Balance

LAYOUT NOTE:
Open/NC pins should have
minimum trace lengths
and must not be
connected to a wire or
cable



Comments:	Company:		Variant:	Git Hash:
			DRAFT	87d29ad
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: Cell Sense & Balance	File Name: cell_sense_and_balance.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Cell Sense & Balance/		Reviewer:	Size: A3	Sheet: 5 of 10

[6] Power & Reference

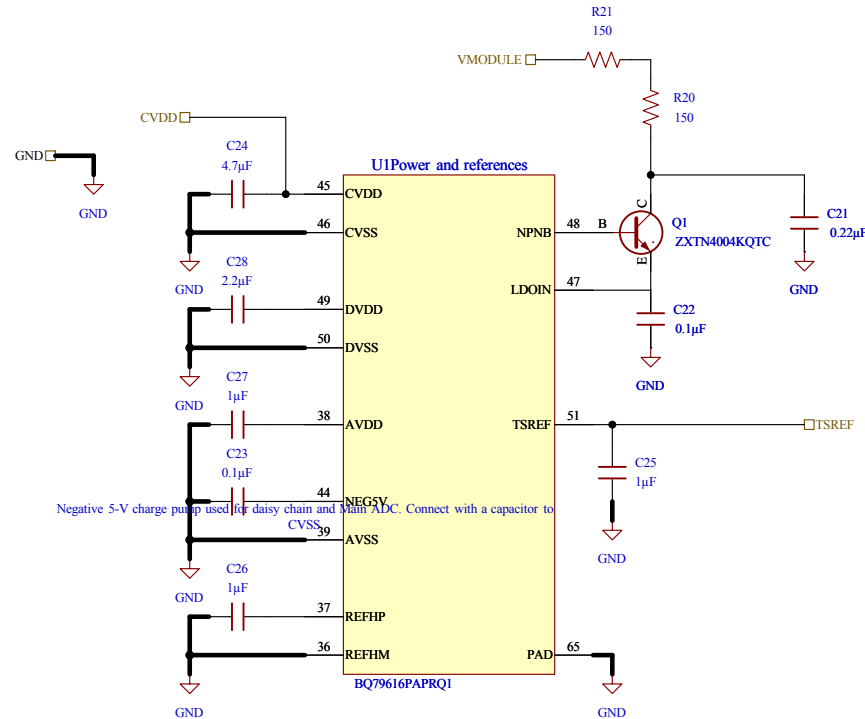
LAYOUT NOTE:
U1 GROUNDING (TI
SLUSE81F §10.1.1)

All grounds are one net (GND) in the schematic. In layout, each bypass cap returns to its own ground pin through a small L1 copper island. Islands join the L2 GND plane through vias only. Do NOT connect islands to each other on L1.

GROUND	PIN	CAPS RETURNING TO IT
AVSS	39	AVDD (38), BAT filter, VCO filter, CB0 filter, thermal pad (65)
DVSS	50	DVDD (49)
CVSS	46	LDOIN (47), NEG5V (44), CVDD (45), TSREF (51), GPIO filter caps, daisy-chain bypass caps
REFHM	36	REFHP (37) ONLY - keep separate from AVSS on L1, join at L2 plane

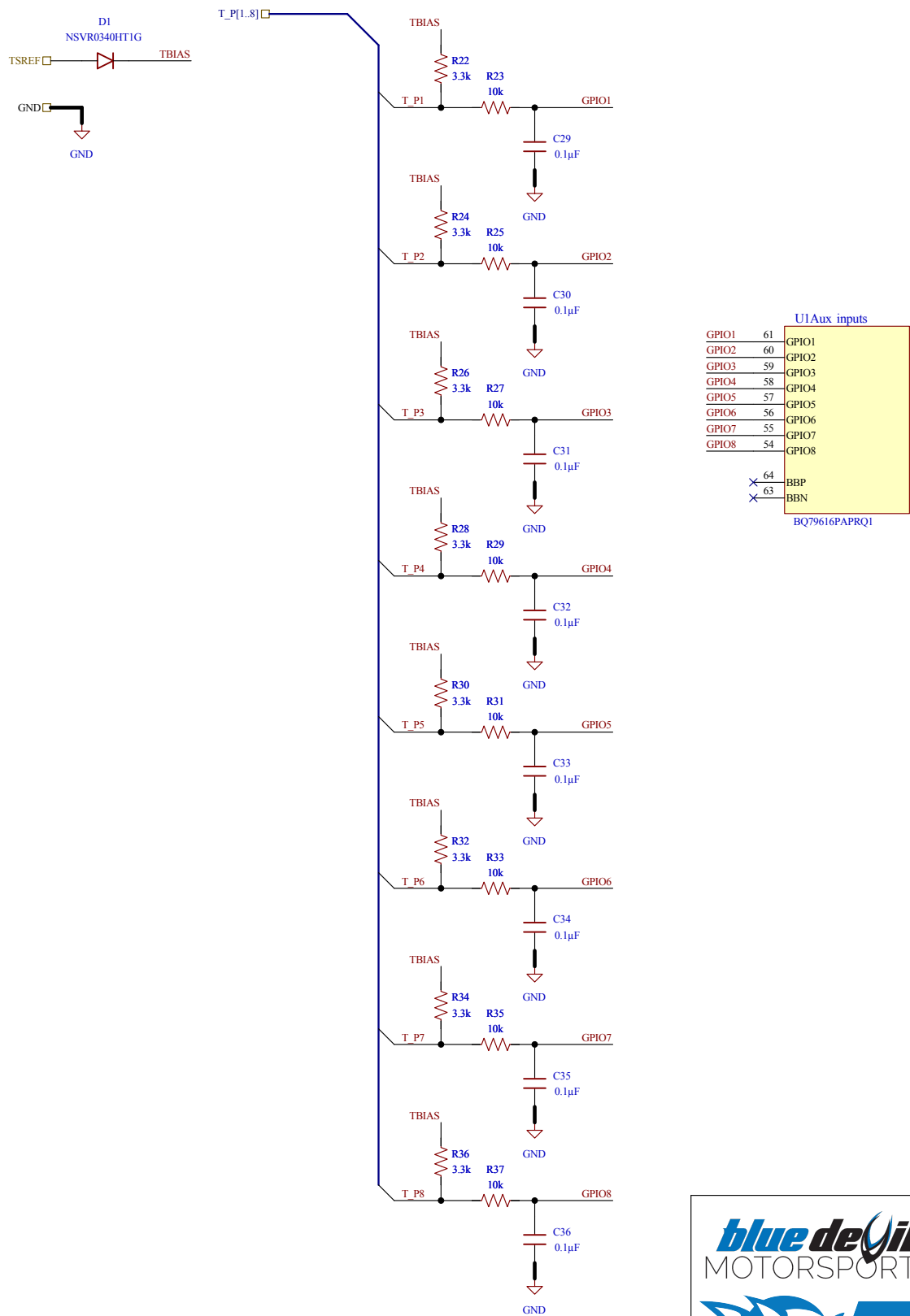
RULES

- Cap within 2 mm of its pin, same layer, no via between cap and pin.
- Cap ground pad => its island => via to L2. Not a shared trace.
- Keep pins 40-43 (COMH/COML) clear, place caps above and below the daisy-chain exit, not in it.
- Thermal pad (65): 36 vias to L2 GND plane, tented/plugged.



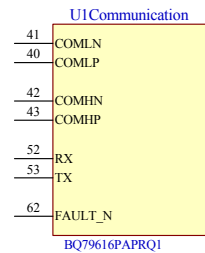
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			DRAFT	87d29ad
Sheet Title: Power & Reference	Board Name: BMS_SLAVE		Project Name: BMS Slave	
	File Name: power_and_reference.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Power & Reference/		Reviewer:	Size: A4	Sheet: 6 of 10

[7] Temperature Sense



Comments:	Company:		Variant:	Git Hash:
	BMS_SLAVE		DRAFT	87d29ad
Sheet Title:	File Name:	Designer:	Date:	Revision:
Temperature Sense	temperature_sense.kicad_sch	Aidan Brzezinski		
Sheet Path:		Reviewer:	Size:	Sheet:
/Project Architecture/Temperature Sense/			A3	7 of 10

[9] Daisy Chain



	Comments:	Company:		Variant:	Git Hash:
				DRAFT	87d29ad
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Daisy Chain	daisy_chain.kicad_sch	Aidan Brzezinski		
	Sheet Path:	Reviewer:	Size:	Sheet:	
	/Project Architecture/Daisy Chain/		A4	9 of 10	

[10] Mechanical

	Comments:		Company:		Variant: DRAFT	Git Hash: 87d29ad
			Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Title: Mechanical		File Name: mechanical.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
	Sheet Path: /Project Architecture/Mechanical/			Reviewer:	Size: A4	Sheet: 10 of 10

[11] Revision History

	Comments:	Company: Blue Devil Motor Sports		Variant: DRAFT	Git Hash: 87d29ad
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 11 of 10